

Least common denominator



1 Find the least common denominator of the following pairs of algebraic fractions:

a $\frac{s}{3}$ and $\frac{s}{9}$

b $\frac{a}{2}$ and $\frac{b}{3}$

c $\frac{1}{18h}$ and $\frac{1}{9h}$

d $\frac{1}{4h}$ and $\frac{1}{4}$

e $\frac{7}{15x^4}$ and $\frac{y}{10x^2}$

f $\frac{1}{x+4}$ and $\frac{1}{x}$

g $\frac{11}{3s}$ and $\frac{17}{5s+54}$

h $\frac{9}{x-7}$ and $\frac{9}{x+7}$

i $\frac{s}{s+3}$ and $\frac{7}{4(s+3)}$

j $\frac{4}{y-8}$ and $\frac{10y}{8-y}$

k $\frac{1}{q(q+5)}$ and $\frac{1}{q(q-2)}$

l $\frac{8}{p+3}$ and $\frac{8}{(p+3)^2}$

m $\frac{1}{(y-7)(y+6)}$ and $\frac{1}{(y-7)(y-4)}$

n $\frac{1}{(b+4)(b-8)}$ and $\frac{1}{(b-3)(b-8)}$

2 Consider the following expression:

$$\frac{3}{x(x+6)} + \frac{10}{(x+6)(x-2)}$$

Find the least common denominator in factored form.

3 Consider the pair of rational expressions:

$$\frac{6}{(m+3)(m+9)} \quad \text{and} \quad \frac{18+4m}{m^2+12m+27}$$

a Factor $m^2 + 12m + 27$.

b Now, find the least common denominator in factored form.

4 Consider the pair of rational expressions:

$$\frac{15+3x}{x+6} \quad \text{and} \quad \frac{11}{x^2+11x+30}$$

a Factor $x^2 + 11x + 30$.

b Now, find the least common denominator in factored form.

5 Consider the pair of rational expressions:



$$\frac{17 + 2x}{3x^2 - 108} \quad \text{and} \quad \frac{9}{x^2 - 36}$$

- a Factor $x^2 - 36$.
- b Factor $3x^2 - 108$.
- c Now, find the least common denominator in factored form.

6 Consider the pair of rational expressions:

$$\frac{3 + 4y}{y^2 + 13y + 40} \quad \text{and} \quad \frac{15}{y^2 + 10y + 16}$$

- a Factor $y^2 + 13y + 40$.
- b Factor $y^2 + 10y + 16$.
- c Now, find the least common denominator in factored form.

7 Consider the pair of rational expressions:

$$\frac{17}{6t^2 + 30t + 36} \quad \text{and} \quad \frac{14 + 3t}{t^2 + 5t + 6}$$

- a Factor $t^2 + 5t + 6$.
- b Factor $6t^2 + 30t + 36$.
- c Now, find the least common denominator in factored form.

Add and subtract rational expressions

8 Consider $\frac{2m}{5}$ and $\frac{4m}{6}$.

- a Find the least common denominator of the two fractions.
- b Now, simplify $\frac{2m}{5} + \frac{4m}{6}$.

9 Fully simplify the following expressions, leaving your answers in factored form:



a $\frac{3x}{10} + \frac{5x+4}{10}$

c $\frac{11x}{14} + \frac{7x}{21}$

e $\frac{8x^2}{11} + \frac{5x^2-2x}{55}$

g $\frac{3x+5}{5} + \frac{5x+4}{3}$

i $\frac{x^2+144}{2(x+12)} + \frac{12x}{x+12}$

b $\frac{x}{5} + \frac{4x}{7}$

d $\frac{7x}{11} + \frac{3x-2}{44}$

f $\frac{x+4}{25} + \frac{3x+4}{10}$

h $\frac{1}{5} + \frac{7}{x-6}$

j $\frac{x+4}{9(x+9)} + \frac{x+4}{x(x+9)}$

10 Fully simplify the following expressions, leaving your answers in factored form:

a $\frac{6x}{2} - \frac{7x}{2}$

c $\frac{3x}{5} - \frac{x}{7}$

e $\frac{4x}{7} - \frac{2x-3}{21}$

g $\frac{5y}{6} - \frac{3y+7}{10}$

i $\frac{5}{8p} - \frac{7}{p}$

k $\frac{x^2}{x-4} - \frac{16}{x-4}$

b $\frac{x+4}{x} + \frac{x+3}{x}$

d $\frac{4x}{45} - \frac{x}{18}$

f $\frac{3x}{10} - \frac{2x+3}{14}$

h $-3x - \frac{3x}{8}$

j $\frac{5x}{7(x+1)} - \frac{10}{x+1}$

l $\frac{1}{x(x-2)} - \frac{5}{(x^2+6)(x-2)}$

11 Fully simplify the following expressions, leaving your answers in factored form:

a $\frac{5}{8y-5} + \frac{10}{7y+7}$

c $\frac{2}{(x+3)(x-10)} + \frac{5}{(x+3)(x+5)}$

e $\frac{x+8}{x^2+19x+88} + \frac{x-7}{(x+11)(x+7)}$

g $\frac{7}{x^2+6x+9} + \frac{4}{5x+15}$

b $\frac{5x-y}{x-y} + \frac{x-6y}{y-x}$

d $\frac{8}{1+w+z} + \frac{7}{9w+9w^2+9wz}$

f $\frac{3}{8x+32} + \frac{2}{x^2+4x}$

h $\frac{4}{x^2-9x+20} + \frac{5}{x^2+4x-32}$

12 Fully simplify the following expressions, leaving your answers in factored form:



a $\frac{x+6}{x+8} - \frac{x-8}{x-6}$

c $\frac{x}{x^2-16} - \frac{12}{x+4}$

e $\frac{3x}{x^2-25} - \frac{2}{(x+3)(x+5)}$

g $\frac{5}{xy-3x+4y-12} - \frac{7}{11x+44}$

b $\frac{x}{2x-6} - \frac{x+5}{3x+21}$

d $\frac{9}{x-4} - \frac{5}{x^2-8x+16}$

f $\frac{4}{x^2-15x+54} - \frac{3}{x^2-36}$

h $\frac{5}{y-2} - \frac{9}{xy-2x-7y+14}$

13 Determine whether each of the following is equivalent to $3 - \frac{x-2}{x+3}$:

a $2 - \frac{1}{x+3}$

b $2 + \frac{5}{x+3}$

c $2 + \frac{1}{x+3}$

d $2 - \frac{5}{x+3}$

14 Determine whether each of the following is equivalent to $\frac{25x^2-2}{5x+1}$:

a $5x-1 + \frac{1}{5x+1}$

b $5x-1 - \frac{1}{5x+1}$

c $5x+1 + \frac{1}{5x+1}$

d $5x+1 - \frac{1}{5x+1}$

Application

15 In an upcoming election, it is anticipated that the number of men who vote, x , will be greater than the number of women who vote, y . Each expression below represents the expected number of people who will not vote in two different counties:

- **County A:** $\frac{x-y}{2}$
- **County B:** $x - \frac{y}{2}$

Which county expects more people to not vote?